

FACTORY FLOW to FORTUNE - PROCESS BOOK

Data Visualizations Final Project (Azad, Sumaiya; Ram, Dhruv; Wood, Joel)

Table of Contents

Basic Info	2
Project Objectives	2
Data Sources	4
Data Processing	5
Configurations	6
Variable Glossary	9
Visualization Design	10
Must-Have Features	19
Optional Features	19
Project Schedule	19

Basic Info

Project Title: Factory Flow to Fortune

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Project Repository: <https://github.com/dataviscourse2025/final-project-factoryflow>

Background and Motivation

This project will be based on principles of operations research and factory physics—disciplines of Industrial Engineering and Management Science. This summer, Joel took ME-EN 6182 (Design of Production and Service Systems), the final project of which involved designing an assembly line system; in this process of designing this system, Joel developed a series of algorithms to give near-optimal configurations for assembly lines—as seen in the feedback given here, these algorithms were able to accomplish the balancing, sequencing, and flow layouts of the assembly line above what was expected for this project. The professor, Pedro Huebner, encouraged additional exploration of these algorithms as they were uniquely developed to handle these practical manufacturing design problems. Reaching out to Pedro, he was enthusiastic in having his original project extended out into a visualization which explored the impacts of ‘design attributes across multiple different configurations.

(+) By far, this is the most comprehensive report I have ever read in 7 years of teaching this course. Kudos for a fantastic execution. Above and beyond. I will save some time to read this with closer attention after I finish grading all 25 reports. I am speechless!

(+) The report was submitted on-time.

(+) The balancing problem was solved and discussed adequately.

(+) The sequencing problem was solved and results were presented.

(+) The proposed design attributes of the line were presented and discussed with clarity.

Assembly line configuration was identified as being particularly well-suited for this visualization project due to having a massive design space, which allows this team a wide range of possible visualizations. There are dozens of key variables which impact assembly lines, and hundreds of minor elements. The inputs and outputs of these are diverse, with many different interactions which are best visualized in different ways. This gives this team a plethora of different visualizations which can be explored both cohesively as a whole, yet also uniquely as individuals. Both inputs and outputs can be communicated statistically, cumulatively, and discretely, both over time and in a moment. With easily dozens of different visualization opportunities which cover all common types, the challenge will be in getting many visualizations to collectively come together, rather than to try and come up with different ways to do so.

Project Objectives

This project addresses a practical problem which arises frequently in a manufacturing environment: how should we design and configure our assembly line? How many people do we need to hire? How much space and what speed of conveyor motors do we need? What is our capacity, and can it match customer demands? And, essential for any business, what combination will allow us to maximize profits? Because the design of a production line is often approved by executive leadership, the ability to quantify and adequately visualize configurations is essential. In this way, development of methods

to find configurations is only part of the problem, as these also need to be communicated. A factory owner will not simply want to see a single configuration, but rather several proposals which can be compared to how they meet the business needs and objectives. Therefore, the development of a tool to explain these through visualizations is critical for establishing confidence in how to allocate capital investments for building a new assembly line.

Additionally, this is a subject which few people taking this course will be familiar with, which allows us to particularly gauge how effectively it communicates these concepts. With so many different interactions, it can be difficult to wrap your head around how so many things influence each other. This means that in addition to being used to communicate business decisions, such a visualization can also be used as an educational tool. This approach will be used here, as other than Joel, the other team members are not familiar with factory physics, capacity planning, or operations research. As such, we plan to build our UI to be exploratory, in such a way that it can be used to learn these principles.

Benefits of such a system, depending on which variables we choose to visualize, would include the ability to answer the following pertinent operational questions:

- Based on expected customer demand, determine the configuration which will generate the most profit, and how many employees will be needed to sustain it?
- Based on an experienced workforce, determine the speed the conveyor should be set to maximize profit and efficiency? Alternatively, based on the wages for employees and the value of the product, how many employees are optimal. How many daily operational hours or shifts are optimal to maximize gross profit margins? If fixed expenses are known, what level of operation is necessary to ensure a net-profit is met?
- Based on set configurations, how much idle time will each workstation have throughout the day? Alternatively, this can be used to allocate employees based on individual experience/efficiency.
- Based on demand and employee count, determine the frequency of outgoing and incoming shipments of raw materials and finished goods. Additionally, determine how much WIP (Work in Progress) will be moving through the factory at a time, and the space constraints of such?
- Based on the number of employees, what order tasks should be completed to maintain precedence requirements, while minimizing idle time and unnecessary movements?
- In a mixed assembly line (where different final products are built with different combinations of core steps) what order should these configurations be built to meet a constant demand, and minimize buildup of WIP and variance between workstations?
- How much floor space will the assembly line take up as you adjust factors such as conveyor speed and spacing of product on that line; how much excess movement will occur as employees move between steps?
- How do the variables of Cycle Time, Throughput, and WIP relate to each other on the assembly line? How do these global variables impact balance delay and efficiency of workstations?

There are numerous other questions which could be informed/answered using visualizations of assembly lines, as they lie at an intersection of resource allocation, financial and operational metrics, and general flow mechanics/physics/statistics within the system. We would like to explore these many interactions, and how to effectively communicate and enable the exploration of these factors to make an educational and informative interface, which enables and encourages its users to take technical design inputs to make meaningful management decisions commonly encountered in manufacturing.

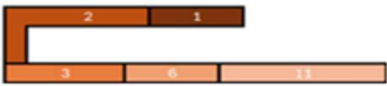
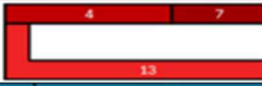
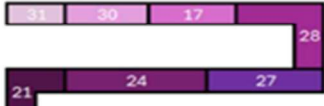
Data Sources

For this project, we will be using the base problem provided by Pedro Huebner, with his expressed permission. This project was based on building an assembly line which would manufacture 3 different models of refrigerators in the following balance: 35% Super, 45% Ultra, and 20% Mega. The assembly process has been broken into a set of discrete elements which can be completed by a single person, so it is a matter of assigning them as needed. Below are the element times used and their predecessor tasks, along with their need based on which model is built. We will have it so total demand can increase or decrease, but the overall composition balance mentioned above will remain the same.

Element	Element Description	Element Time (min)	Predecessor	Required?		
				Super	Ultra	Mega
1	Place frame on holder and secure	0.8	-	1	1	1
2	Install internal heat exchanging pipes (IHEPs)	1.5	1	1	1	1
3	Install compressor motor/pump kit	1.0	1	1	1	1
4	Install compressor electrical kit	1.3	3	1	1	1
5	Connect IHEPs to compressor	0.5	2, 3	1	1	1
6	Install defrost heater kit	0.7	1	1	1	1
7	Install defrost drain tube	0.7	6	1	1	1
8	Install water filter kit	0.8	1	0	1	1
9	Connect and route water pipes	0.5	8	0	1	1
10	Install basic electronic control board	1.2	1	1	1	0
11	Install advanced electronic control board	1.5	1	0	0	1
12	Connect and route electrical wires	2.0	10, 11	1	1	1
13	Install internal panels	2.5	4, 5, 7, 9, 12	1	1	1
14	Install evaporator and evaporator fans	1.5	13	1	1	1
15	Install external panels	2.2	14	1	1	1
16	Install external heat exchanging pipes (EHEPs)	1.2	15	1	1	1
17	Connect EHEPs to compressor	0.5	16	1	1	1
18	Install door brackets	0.8	14	1	1	1
19	Install freezer compartment door	1.1	18	1	1	1
20	Install icemaker kit into freezer door	0.5	19	0	1	1
21	Connect pre-routed water pipes to icemaker	0.5	20	0	1	1
22	Install refrigerator compartment door	1.3	18	1	1	1
23	Install touchscreen (TS) into refrigerator door	0.7	22	0	0	1
24	Connect pre-routed electrical wires to TS	0.9	23	0	1	1
25	Install sensors and switches	1.3	19, 22	1	1	1
26	Install lighting kit	1.0	19, 22	1	1	1
27	Connect remaining pre-routed electrical wires	0.7	25, 26	1	1	1
28	Assemble internal shelves and drawers	1.0	27	1	1	1
29	Install feet and wheels	0.7	15	1	1	1
30	Visual and structural inspection	0.5	17, 21, 24, 27, 29	1	1	1
31	Remove from holder and package	0.4	30	1	1	1

Data Processing

While it may be possible to rewrite the algorithms for the original project (used for balancing, sequencing, and optimizing flow) to function within JavaScript, this will greatly expand the complexity of the backend. Therefore, to keep the focus of this project on developing front-end visualizations, we will not have these calculations be performed dynamically within our project. It has already been determined that more than 12 employees will not represent any form of optimization across any possible metric, due to discrete element 13 making up 8.7% of the total task demand time. We will therefore pre-emptively calculate near-optimal configurations for between 3 to 12 employees, with our established model build ratios. These will generate unique sequences for the 31 elements, which will be used in conjunction with their ratios of total daily time, as shown in the below 7-workstation configuration diagram.

Element	Task Time (m)	Daily Time (m)	Station Length (ft)	Station Efficiency (%)	Counter Flow Walking Time (m)	Adjusted Daily Time (m)
WorkStation 1	6.1	860	91.5	95.6%	7.44	868
1	0.8	160	12			
2	1.5	300	22.5			
3	1	200	15			
6	1.3	140	19.5			
11	1.5	60	22.5			
WorkStation 2	5	861	75	95.7%	20.24	882
10	1.2	192	18			
5	0.5	100	7.5			
8	0.8	104	12			
9	0.5	65	7.5			
12	2	400	30			
WorkStation 3	4.5	900	67.5	100.0%	4.80	905
7	0.7	140	10.5			
4	1.3	260	19.5			
13	2.5	500	37.5			
WorkStation 4	3.7	740	55.5	82.2%	5.36	746
14	1.5	300	22.5			
15	2.2	440	33			
WorkStation 5	4	800	60	88.9%	9.60	810
18	0.8	160	12			
22	1.3	260	19.5			
16	1.2	240	18			
29	0.7	140	10.5			
WorkStation 6	4.6	773	69	85.9%	12.57	786
19	1.1	220	16.5			
26	1	200	15			
20	0.5	65	7.5			
23	0.7	28	10.5			
25	1.3	260	19.5			
WorkStation 7	4.5	802	67.5	89.1%	9.60	812
21	0.5	65	7.5			
24	0.9	117	13.5			
27	0.7	140	10.5			
28	1	200	15			
17	0.5	100	7.5			
30	0.5	100	7.5			
31	0.4	80	6			
TOTAL	32.4 min	5736 min	486 ft		71 min	5809 min
Floor Space	Throughput	Cycle Time	Conveyor Speed	Product Spacing	Balance Delay	Efficiency
57' X 90'	13.33 pph	4.5 min	15 ft/min	67.5 ft	8.95%	91.05%

These pre—established elements will then be used to dynamically calculate our needed variables for our visualizations; these pre-calculations are outlined below, within “Configurations”.

Configurations

3 Workers

WorkStation 1: Task 1
WorkStation 1: Task 2
WorkStation 1: Task 3
WorkStation 1: Task 10
WorkStation 1: Task 11
WorkStation 1: Task 12
WorkStation 1: Task 4
WorkStation 1: Task 5
WorkStation 1: Task 6
WorkStation 1: Task 8

WorkStation 2: Task 9
WorkStation 2: Task 7
WorkStation 2: Task 13
WorkStation 2: Task 14
WorkStation 2: Task 18
WorkStation 2: Task 19
WorkStation 2: Task 22
WorkStation 2: Task 25

WorkStation 3: Task 20
WorkStation 3: Task 21
WorkStation 3: Task 15
WorkStation 3: Task 16
WorkStation 3: Task 26
WorkStation 3: Task 23
WorkStation 3: Task 24
WorkStation 3: Task 27
WorkStation 3: Task 28
WorkStation 3: Task 29
WorkStation 3: Task 17
WorkStation 3: Task 30
WorkStation 3: Task 31

4 Workers

WorkStation 1: Task 1
WorkStation 1: Task 2
WorkStation 1: Task 3
WorkStation 1: Task 10
WorkStation 1: Task 11
WorkStation 1: Task 12
WorkStation 1: Task 6

WorkStation 2: Task 8
WorkStation 2: Task 9
WorkStation 2: Task 4
WorkStation 2: Task 5
WorkStation 2: Task 7
WorkStation 2: Task 13
WorkStation 2: Task 14

WorkStation 3: Task 15
WorkStation 3: Task 18
WorkStation 3: Task 19
WorkStation 3: Task 22
WorkStation 3: Task 25
WorkStation 3: Task 20

WorkStation 4: Task 21
WorkStation 4: Task 16
WorkStation 4: Task 26
WorkStation 4: Task 27
WorkStation 4: Task 23
WorkStation 4: Task 24
WorkStation 4: Task 17
WorkStation 4: Task 28
WorkStation 4: Task 29
WorkStation 4: Task 30
WorkStation 4: Task 31

5 Workers

WorkStation 1: Task 1
WorkStation 1: Task 3
WorkStation 1: Task 10
WorkStation 1: Task 2
WorkStation 1: Task 5
WorkStation 1: Task 8

WorkStation 2: Task 9
WorkStation 2: Task 4
WorkStation 2: Task 6
WorkStation 2: Task 11
WorkStation 2: Task 7
WorkStation 2: Task 12

WorkStation 3: Task 13
WorkStation 3: Task 14
WorkStation 3: Task 18
WorkStation 3: Task 19

WorkStation 4: Task 15
WorkStation 4: Task 16
WorkStation 4: Task 22
WorkStation 4: Task 26
WorkStation 4: Task 20

WorkStation 5: Task 23
WorkStation 5: Task 21
WorkStation 5: Task 24
WorkStation 5: Task 17
WorkStation 5: Task 25
WorkStation 5: Task 27
WorkStation 5: Task 28
WorkStation 5: Task 29
WorkStation 5: Task 30
WorkStation 5: Task 31

6 Workers

WorkStation 1: Task 1
WorkStation 1: Task 2
WorkStation 1: Task 3
WorkStation 1: Task 5
WorkStation 1: Task 6
WorkStation 1: Task 11

WorkStation 2: Task 4
WorkStation 2: Task 7
WorkStation 2: Task 10
WorkStation 2: Task 12

WorkStation 3: Task 8
WorkStation 3: Task 9
WorkStation 3: Task 13
WorkStation 3: Task 14

WorkStation 4: Task 18
WorkStation 4: Task 19
WorkStation 4: Task 22
WorkStation 4: Task 26
WorkStation 4: Task 20
WorkStation 4: Task 23

WorkStation 5: Task 15
WorkStation 5: Task 16
WorkStation 5: Task 25

WorkStation 6: Task 21
WorkStation 6: Task 24
WorkStation 6: Task 17
WorkStation 6: Task 27
WorkStation 6: Task 28
WorkStation 6: Task 29
WorkStation 6: Task 30
WorkStation 6: Task 31

7 Workers

Workstation 1: Task 1
Workstation 1: Task 2
Workstation 1: Task 3
Workstation 1: Task 6
Workstation 1: Task 11

Workstation 2: Task 10
Workstation 2: Task 12
Workstation 2: Task 5
Workstation 2: Task 8
Workstation 2: Task 9

Workstation 3: Task 4
Workstation 3: Task 7
Workstation 3: Task 13

Workstation 4: Task 14
Workstation 4: Task 15

Workstation 5: Task 16
Workstation 5: Task 18
Workstation 5: Task 22
Workstation 5: Task 29

Workstation 6: Task 19
Workstation 6: Task 25
Workstation 6: Task 26
Workstation 6: Task 20
Workstation 6: Task 23

Workstation 7: Task 26
Workstation 7: Task 27
Workstation 7: Task 21
Workstation 7: Task 24
Workstation 7: Task 17
Workstation 7: Task 30
Workstation 7: Task 31

8 Workers

WorkStation 1: Task 1
WorkStation 1: Task 2
WorkStation 1: Task 10

WorkStation 2: Task 11
WorkStation 2: Task 12
WorkStation 2: Task 3
WorkStation 2: Task 6

WorkStation 3: Task 8
WorkStation 3: Task 4
WorkStation 3: Task 7
WorkStation 3: Task 5
WorkStation 3: Task 9

WorkStation 4: Task 13
WorkStation 4: Task 14

WorkStation 5: Task 18
WorkStation 5: Task 22
WorkStation 5: Task 19

WorkStation 6: Task 15
WorkStation 6: Task 25

WorkStation 7: Task 26
WorkStation 7: Task 16
WorkStation 7: Task 27
WorkStation 7: Task 29

WorkStation 8: Task 20
WorkStation 8: Task 21
WorkStation 8: Task 28
WorkStation 8: Task 23
WorkStation 8: Task 24
WorkStation 8: Task 17
WorkStation 8: Task 30
WorkStation 8: Task 31

9 Workers

WorkStation 1: Task 1
WorkStation 1: Task 2
WorkStation 1: Task 10

WorkStation 2: Task 11
WorkStation 2: Task 3
WorkStation 2: Task 4
WorkStation 2: Task 6

WorkStation 3: Task 8
WorkStation 3: Task 7
WorkStation 3: Task 12

WorkStation 4: Task 9
WorkStation 4: Task 5
WorkStation 4: Task 13

WorkStation 5: Task 14
WorkStation 5: Task 18

WorkStation 6: Task 15
WorkStation 6: Task 19

WorkStation 7: Task 22
WorkStation 7: Task 25
WorkStation 7: Task 29

WorkStation 8: Task 20
WorkStation 8: Task 16
WorkStation 8: Task 26
WorkStation 8: Task 23
WorkStation 8: Task 24

WorkStation 9: Task 21
WorkStation 9: Task 17
WorkStation 9: Task 27
WorkStation 9: Task 28
WorkStation 9: Task 30
WorkStation 9: Task 31

10 Workers

WorkStation 1: Task 1
WorkStation 1: Task 2
WorkStation 1: Task 10

WorkStation 2: Task 11
WorkStation 2: Task 3
WorkStation 2: Task 4
WorkStation 2: Task 6

WorkStation 3: Task 8
WorkStation 3: Task 7
WorkStation 3: Task 12

WorkStation 4: Task 9
WorkStation 4: Task 5
WorkStation 4: Task 13

WorkStation 5: Task 14
WorkStation 5: Task 18

WorkStation 6: Task 15
WorkStation 6: Task 29

WorkStation 7: Task 22
WorkStation 7: Task 23
WorkStation 7: Task 19

WorkStation 8: Task 25
WorkStation 8: Task 20
WorkStation 8: Task 26

WorkStation 9: Task 16
WorkStation 9: Task 21
WorkStation 9: Task 24
WorkStation 9: Task 17

WorkStation 10: Task 27
WorkStation 10: Task 28
WorkStation 10: Task 30
WorkStation 10: Task 31

11 Workers

WorkStation 1: Task 1
WorkStation 1: Task 3
WorkStation 1: Task 8
WorkStation 1: Task 9

WorkStation 2: Task 2
WorkStation 2: Task 10
WorkStation 2: Task 11

WorkStation 3: Task 5
WorkStation 3: Task 12

WorkStation 4: Task 4
WorkStation 4: Task 6
WorkStation 4: Task 7

WorkStation 5: Task 13

WorkStation 6: Task 14
WorkStation 6: Task 18

WorkStation 7: Task 22
WorkStation 7: Task 23
WorkStation 7: Task 19

WorkStation 8: Task 15
WorkStation 8: Task 20

WorkStation 9: Task 16
WorkStation 9: Task 17
WorkStation 9: Task 26

WorkStation 10: Task 25
WorkStation 10: Task 27
WorkStation 10: Task 29

WorkStation 11: Task 21
WorkStation 11: Task 24
WorkStation 11: Task 28
WorkStation 11: Task 30
WorkStation 11: Task 31

12 Workers

WorkStation 1: Task 1
WorkStation 1: Task 3
WorkStation 1: Task 8
WorkStation 1: Task 9

WorkStation 2: Task 10
WorkStation 2: Task 2
WorkStation 2: Task 11

WorkStation 3: Task 12
WorkStation 3: Task 5

WorkStation 4: Task 4
WorkStation 4: Task 6
WorkStation 4: Task 7

WorkStation 5: Task 13

WorkStation 6: Task 14

WorkStation 7: Task 18
WorkStation 7: Task 22
WorkStation 7: Task 23

WorkStation 8: Task 15

WorkStation 9: Task 19
WorkStation 9: Task 25

WorkStation 10: Task 26
WorkStation 10: Task 27
WorkStation 10: Task 29

WorkStation 11: Task 20
WorkStation 11: Task 16
WorkStation 11: Task 17
WorkStation 11: Task 21

WorkStation 12: Task 28
WorkStation 12: Task 24
WorkStation 12: Task 30
WorkStation 12: Task 31

Element

Percentage of Total

(Labor Time/Conveyor Length)

Element 1 - 2.79% / 2.52%
Element 2 - 5.23% / 4.72%
Element 3 - 3.49% / 3.15%
Element 4 - 4.53% / 4.09%
Element 5 - 1.74% / 1.57%
Element 6 - 2.44% / 2.20%
Element 7 - 2.44% / 2.20%
Element 8 - 1.81% / 2.52%
Element 9 - 1.13% / 1.57%
Element 10 - 3.35% / 3.77%
Element 11 - 1.05% / 4.72%
Element 12 - 6.97% / 6.29%
Element 13 - 8.72% / 7.86%
Element 14 - 5.23% / 4.72%
Element 15 - 7.67% / 6.92%
Element 16 - 4.19% / 3.77%
Element 17 - 1.74% / 1.57%
Element 18 - 2.79% / 2.52%
Element 19 - 3.84% / 3.46%
Element 20 - 1.13% / 1.57%
Element 21 - 1.13% / 1.57%
Element 22 - 4.53% / 4.09%
Element 23 - 0.49% / 2.20%
Element 24 - 2.04% / 2.83%
Element 25 - 4.53% / 4.09%
Element 26 - 3.49% / 3.14%
Element 27 - 2.44% / 2.20%
Element 28 - 3.49% / 3.14%
Element 29 - 2.44% / 2.20%
Element 30 - 1.74% / 1.57%
Element 31 - 1.40% / 1.26%

Build Sequence

1 (Super) 2 (Ultra) 3 (Mega)

2 1 3 2 1 2 1 3 2 1 2 2 3
1 2 1 2 3 1 2 2 1 3 2 1 2
1 3 2 2 1 2 3 1 2 1 2 3 1
2 2 1 3 2 1 2 1 3 2 1 2 2
3 1 2 1 2 3 1 2 2 1 3 2 1
2 1 3 2 1 2 2 3 1 2 1 2 3
1 2 2 1 3 2 1 2 1 3 2 1 2
2 3 1 2 1 2 3 1 2 2 1 3 2
1 2 1 3 2 1 2 2 3 1 2 1 2
3 1 2 2 1 3 2 1 2 1 3 2 1
2 2 3 1 2 1 2 3 1 2 2 1 3
2 1 2 1 3 2 1 2 2 3 1 2 1
2 3 1 2 2 1 3 2 1 2 1 3 2
1 2 2 3 1 2 1 2 3 1 2 2 1
3 2 1 2 1 3 2 1 2 2 3 1 2
1 2 3 1 2

Bayesian Flow Weights

Super → Super: 0%
Super → Ultra: 71.43%
Super → Mega: 28.57%

Ultra → Super: 56.18%
Ultra → Ultra: 21.35%
Ultra → Mega: 22.47%

Mega → Super: 50%
Mega → Ultra: 0%
Mega → Mega: 50%

Variable Glossary

Mixed Assembly Line – a form of manufacturing where building a product is subdivided into different tasks to collectively form the finished product. Products are moved steps by a conveyor system which moves at a fixed speed. Mixed Lines will build several different variations of a product which share common steps. Some steps may or may not be needed for a specific model.

Workstation – series of tasks (elements) intended to be performed by an individual, sequentially.

Elements – These are atomic tasks which are done by a single person at a time; they cannot be shared between people or workstations; however, they may be performed sequentially.

Precedence – When building a product, certain steps may need to be completed before other steps can begin, these steps which must be completed are known as predecessor elements.

Element Time – how long tasks take to complete once, from beginning to end. This with conveyor properties (spacing and speed), directly influences the physical length of the conveyor needed for the task.

Task/Labor Time – Represents the amount of daily time an individual is expected to perform the task within a given timeframe. Within a mixed assembly line, this time will not be the same as the element time, as a process will always take the same amount of space/time, but what it takes up within a day will be based on the frequency they are built.

WIP – Work in Progress, the number of incomplete products currently being worked on.

Flow – the visual depiction of WIP moving through the assembly line, with a steady movement rate.

Demand – the number of units which can be sold, or which producing more than needed is a liability. Production should match your demand.

Throughput – the number of models built within a given time-period, usually an hour.

Workstation Time – the sum of all Task times in a workstation, representing how much time an operator will spend building one model through their station. Directly impacts conveyor lengths.

Cycle Time – the average amount of time which passes between each completed model to come off the assembly line.

Process Time – the total time of all element times, representing the amount of time which passes for a model to move through the whole line.

Efficiency – percentage of time in which the assembly line/workstation are working on a model.

Balance Delay – the percentage of time in which the individual(s) in a workstation/line are idle / not working.

BD CV – The variance of balance delay between workstations normalized by their average balance delay. Used to evaluate how evenly distributed tasks are between workstations/individuals.

COGS – Cost of Goods Sold, cost of materials & labor to produce an item. Generally, it represents variable costs which fluctuate by how many goods are sold.

Gross Profit – The value of goods sold minus COGS. Representing the value a given product provides for the company, to sustain operations.

Net Profit – Gross Profit minus Operating Expenses (generally fixed costs, which are incurred regardless of production), representing final profit of the company.

Marginal Profit – the per unit value that a product line provides in gross profit.

Profit Margin – can be either net or gross, this is the percentage of profit compared to total goods sold.

Set Variables

(These are preset and cannot be changed without building up an extensive back end)

Build Composition (the percentage of each type of model built of total demand), Conveyor Distance (the distance between each item on the conveyor belt), Precedence, Element Time.

Adjustable Variables

(These have preset values which can be adjusted to change outputs)

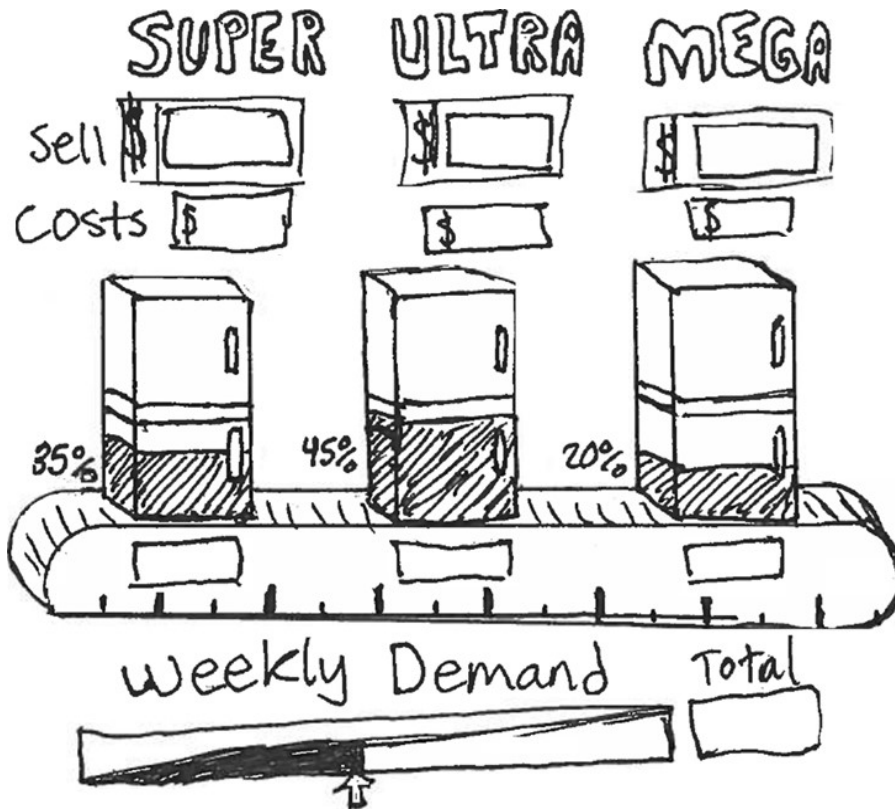
Demand, Labor Costs (per hour), Operational Hours (How many hours a day the assembly line is running), Sell Price and Cost of Goods (per model), Workstation Compositions (which elements are in workstations), Number of Employees,

Output Variables

(These are shown as output characteristics of the line, but aren't directly adjustable)

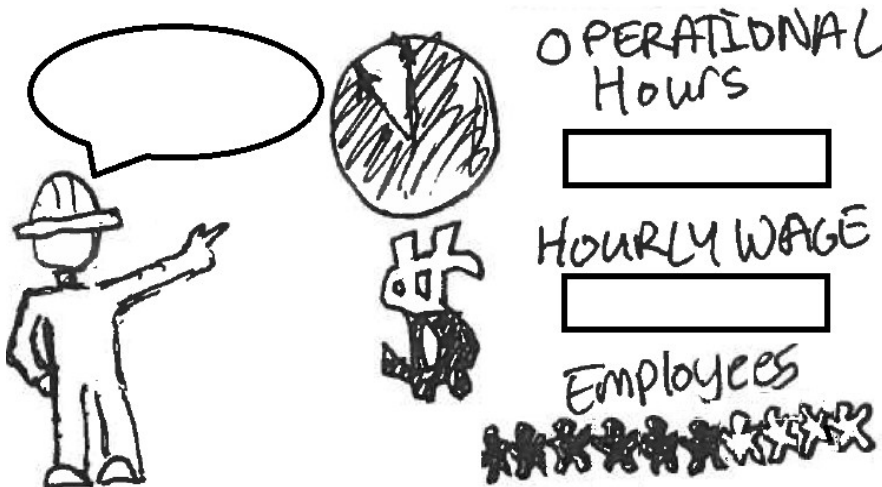
Conveyor Speed, Task/Labor Times, WIP, Throughput, Cycle Time, Process Time, Efficiency, Balance Delay, BD CV, All Financial Values (COGS, Profits, Marginal and Margin)

Visualization Design



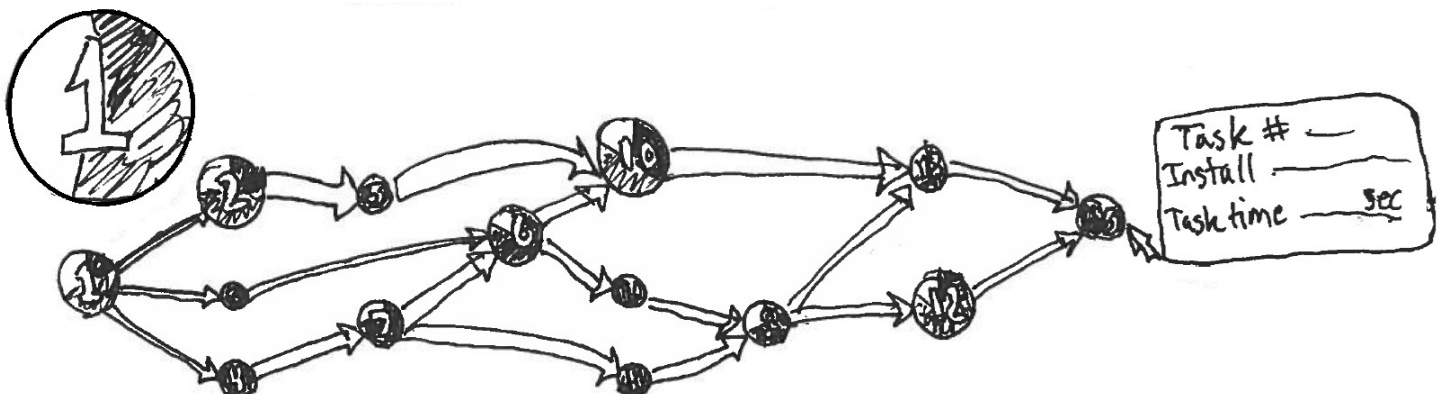
ADJUSTABLE INPUTS #1

A conveyor belt image is used to show the 3 different models of refrigerators. Each of the three models is shaded in a percentage corresponding to its build ratio percentage of the total demand. These total numbers are shown in boxes below each fridge, as an output field. Below there is a slider and a text field which can both set the total weekly demand. Above is the title of the three fridges, with their total Sell cost for each model, and their material costs below. These will be prefilled, but adjustable through text. Finally, a ruler will be shown as the base of the conveyor, showing the distance that the Fridges are, and the distance between them.

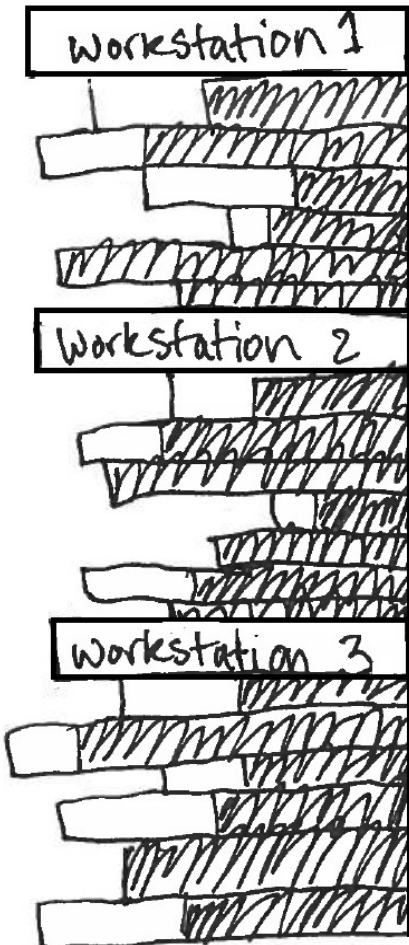


Factory Worker inputs are shown in the form of a clock which fills up based on the number of hours worked in a day, and a dollar sign which behaves similarly. These are input in fields to the right. A meeple fill-in shows the number of recommended / changed employees. A worker on the side has a textbox to give feedback on input selections. Maybe they show their satisfaction with their expression?

SET VARIABLES #2



Here, the elements of assembly are shown in a precedence DAG where the nodes represent the individual elements of assembly. These are represented with their element number overlaid over a pie-chart which shows what percentage of their time is spent on the 3 different models (preferably with the same color as the fridges in the above diagram, where each model uses a different color). The size of the node will vary based on the amount of time each day an operator will need to be working on that element. Hovering over an element will display a tooltip which lists the element number, a description of the task, and how long that task is expected to take.



The edges of the DAG convey successor tasks, with the thickness of the arrows representing the element time between the tasks. This will be directly impactful on the length of the conveyor, since conveyor speed will be constant throughout the whole process. This section is purely informative, and is not intended to change at all, so much as to convey critical information about what steps are needed to build the mix of different refrigerators.

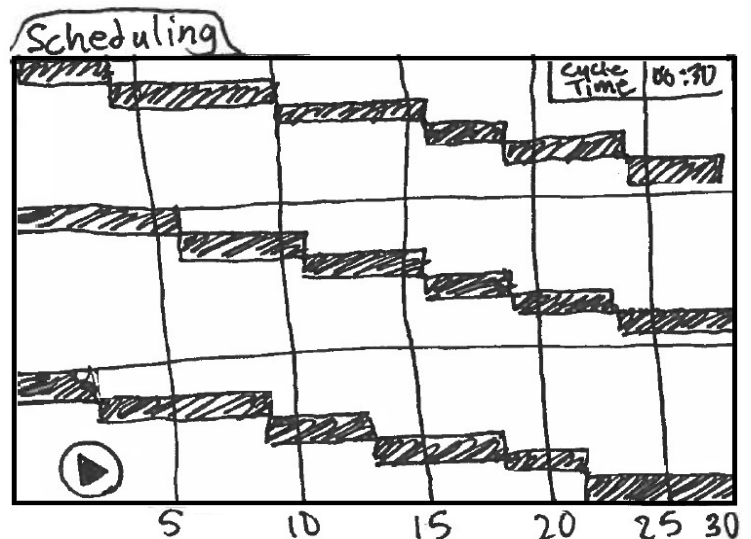
OUTPUTS #3

To the left of the various output panes, there is a ribbon which shows the sequence of the different elements, as they are allocated to workstations/employees. These are useful to actively show, as the sequencing and allocation is fundamental, along with demand, in influencing almost all outputs. If possible, allowing for a drag and drop of elements between Workstations could allow more dynamic exploration. Beneath each Workstation tab, the elements are listed in sequence as they would be performed in the workstation. The length of the total bars represents element times, and in turn how long their conveyor segments would be. Within these, the bars are shaded with what percentage of the time they are being built. The balance of tasks will be based on getting all workstations to have a similar total length of shaded bars, while meeting precedence requirements.

For convenience, it would be good to have a retractable right ribbon for the various ADJUSTABLE INPUTS, so that they can be modified in real-time, without needing to scroll up and down. For the main output area, a series of “folder tabs” can be used to quickly switch between different views which will show various behaviors of both workstations and the whole assembly line in real time. These tabs will be viewed individually as a subset of useful data which collectively tell a story.

SCHEDULING

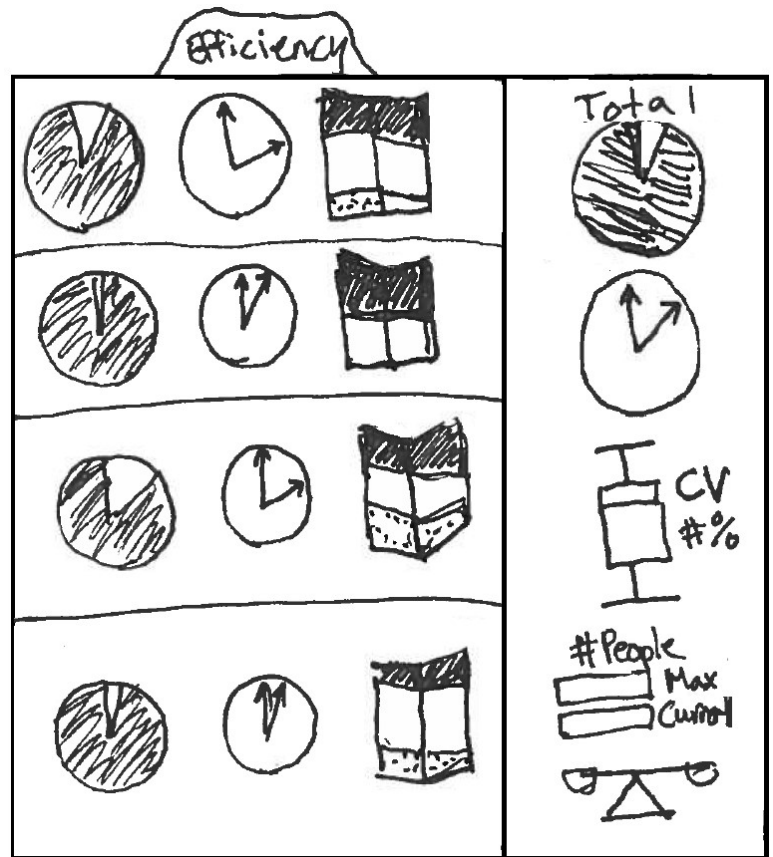
For this, we can use a Gantt Chart in one of two ways. One is as a simulation in which a “Play” button is pressed, and the various tasks as their full (unshaded) bars will flow across sequentially in building a given model—which follows the



predetermined optimal sequence. This would show which areas different employees would be working at a given time. Alternatively, the shaded bars which represent total daily time could also be arranged as a Gantt chart, showing the overall sequencing of tasks. This would not be a true representation of task times and would need to be shown as a portion of the day. This can simply be used as a visual aid of how different tasks contribute to overall labor in workstations.

EFFICIENCY

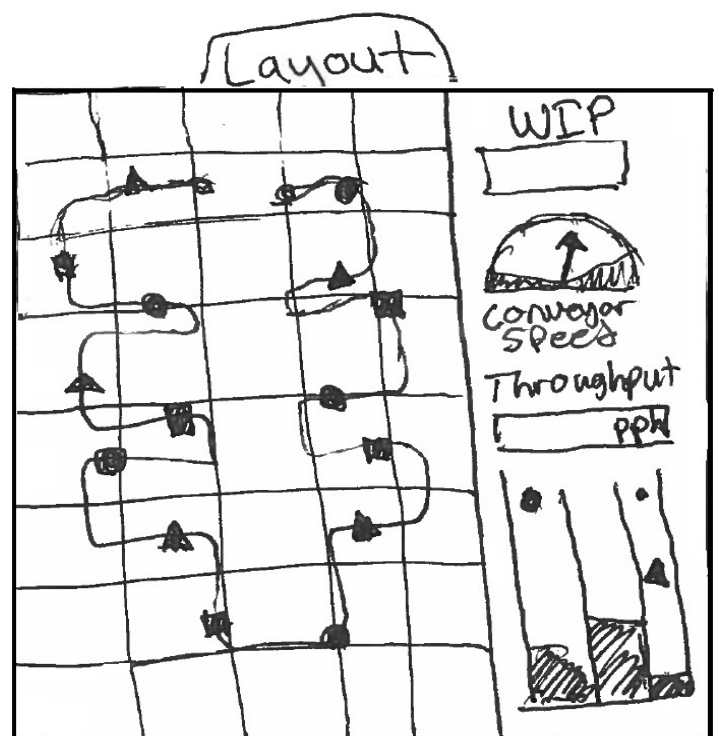
This one can focus on the behavior of individual workstations, with a series of area diagrams. The first, a pie-chart, will show a percentage of efficiency, filled with what percentage of the time an employee is working. Next to it will be a clock which shows how much time is spent idle at a given station throughout the operational day. This is an absolute value, compared to the relative value given by efficiency. Finally, a 3D stacked bar chart is used to connect the concept for how long individual models take to build at the station (the sum of their relevant tasks) and how those extend to a total amount of time spent each day on those tasks/model. A tooltip can hover over each of these graphical representations to show the details, for the clock, it could show the per annum cost of the idle time in employee wages.



To the far side of the tab, a segment should show the full assembly line. A larger version of the pie chart and clock, showing total impact of non-applied time could be shown. Additionally, a Box and Whisker Plot could be used to show the standard deviation of the workstations, with a scale to the side showing a coefficient of variance. Below this, there is a text box showing what the current number of workstations/ employees are, and what the current and maximum output those employees can perform. This can either be based on demand or throughput. Below this, a visual scale can show the balance of the line between being overloaded or being too idle. This can give a visual for how optimal the current configuration is for the given workstations.

LAYOUT

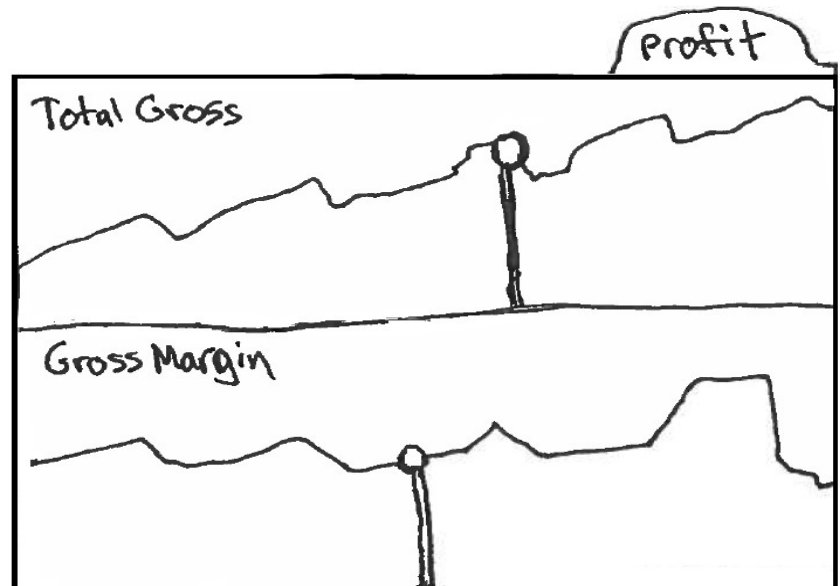
A flow chart overlaid over a floor grid, this could show the assembly line as both different colored workstations, and varied saturation elements. On the line, the different models could be seen flowing



along the line, with a different color/shape for each of the models. The shapes could shift between being outlined and filled based on when the element they are passing through is relevant for them. As the models reach the end of the line, they could “fall” or bounce into different buckets of finished goods, each one for a different model. Showing the accumulation of demand over time. Finally, a Counter box could show WIP (the units moving through at a given time), with a speedometer below showing the conveyor speed. Below with the finished goods, there can be a display for the throughput, that is, the number of finished goods per hour.

PROFIT

Two main metrics should be considered for a selection of gross profit, the total and the margin. While marginal profit (the value generated by an additional item) may be of interest, it is likely to get obfuscated alongside profit margin for anyone who is not accustomed to financial analysis. The use of marginal profit is useful in a microeconomic analysis of what value an expansion in demand is in the market. This may be something we



can find a good fit in showing, perhaps if we do include some market elements. But it is not a good fit in the current layout, and unless market analysis becomes part of our scope, this is unlikely to change. It is absolutely a valuable metric for determining when production should be increased to a higher demand. But the scope of this is to function as a factory flow simulator, and not a full business decision making simulator. The effect of understanding when demand should shift can be understood by evaluating the total and margin. In the above chart, to show profit, we can have a line/area chart which shows how profit behaves as demand is increased based on our calculated near-optimal configurations. There will be a vertical line which shows what demand is currently set at, with a point which can move up and down to indicate the current profit generated in the configuration. A Tooltip can be used over the line to show optimal configurations to get that profit and hovering over the point will show the current configuration. This will help conceptualize the financial impact that these configuration flows have.

LAYOUTS/ALTERNATIVES

It was also suggested that having a set of different pages may be useful, such as a landing page which would introduce the problem space and introduce the team, as well as a dropdown to navigate between the pages.

This could perhaps become a page for initial variables and the PERT chart showed above.

Factory Flow To Fortune

Abstract/About

Dropdown Pages
Page 1
Page 2
Page 3

Landing Page

Factory Flow To Fortune

Build Your Army	Number of Employees Enter #	Operational Hours Enter #	Labor Costs/Person/hr Enter #
Build Composition	Super Fridges 35%	Ultra Fridges 45%	Mega Fridges 20%
Misc.	Demand Enter #	Conveyor Distance Set Value	Sell Price & Cost of Goods Enter #

Dropdown Pages

Page 1

Page 2

Build Your Army

Build Composition

Misc.

Element Time

Precedence (via PERT)

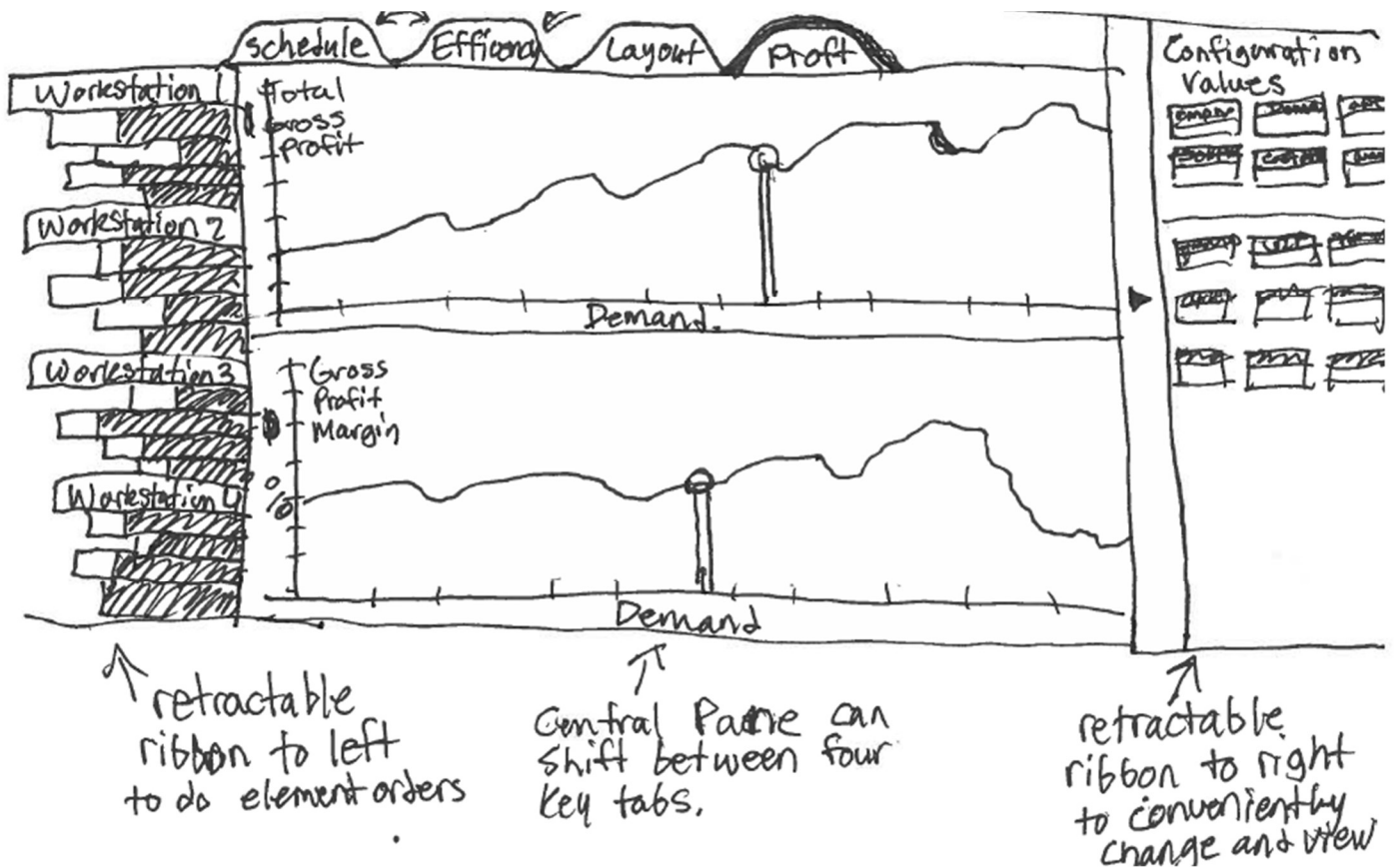
Workstation Composition

Output Variables

Assembly Line

Page 3

Perhaps showing too many visualizations for the inputs will just overly complicate the interface and deter users while unnecessarily expanding scope. It was suggested that having a panel which allowed variables to be set and viewed just as numbers may be easier for some users to interface with than a wide range of different visualized inputs. Initially this was proposed as a subpage, but this may involve too many screen changes for people looking to see multiple views dynamically. It was therefore proposed to make a right ribbon which can be expanded or collapsed with this view, as seen below.



FINAL PROPOSED TEMPLATING

Across several different templates designs, the following was identified as an optimal way to minimize scrolling while viewing dynamic systems, reducing unnecessary over-visualization, and allowing the problem space to be well defined and communicated. The website will be divided between two pages, a primary landing page which defines the problem space, explains set configurations, introduces key adjustable variables and why they matter, and allows initial values to be set for the main visualization.

Of note, Sell Price and Material Costs will have a value for all three models. These will make up the key adjustable variables needed to generate the following dashboard. However, sometimes the operational hours make it impossible for demand to be met, in this case, the lower bounds for these will change, and an Error will show in the boxes. An upper limit will similarly be set on Demand, as no more than 576 units can be produced in a day, due to the greatest atomic element taking 2.5 minutes.

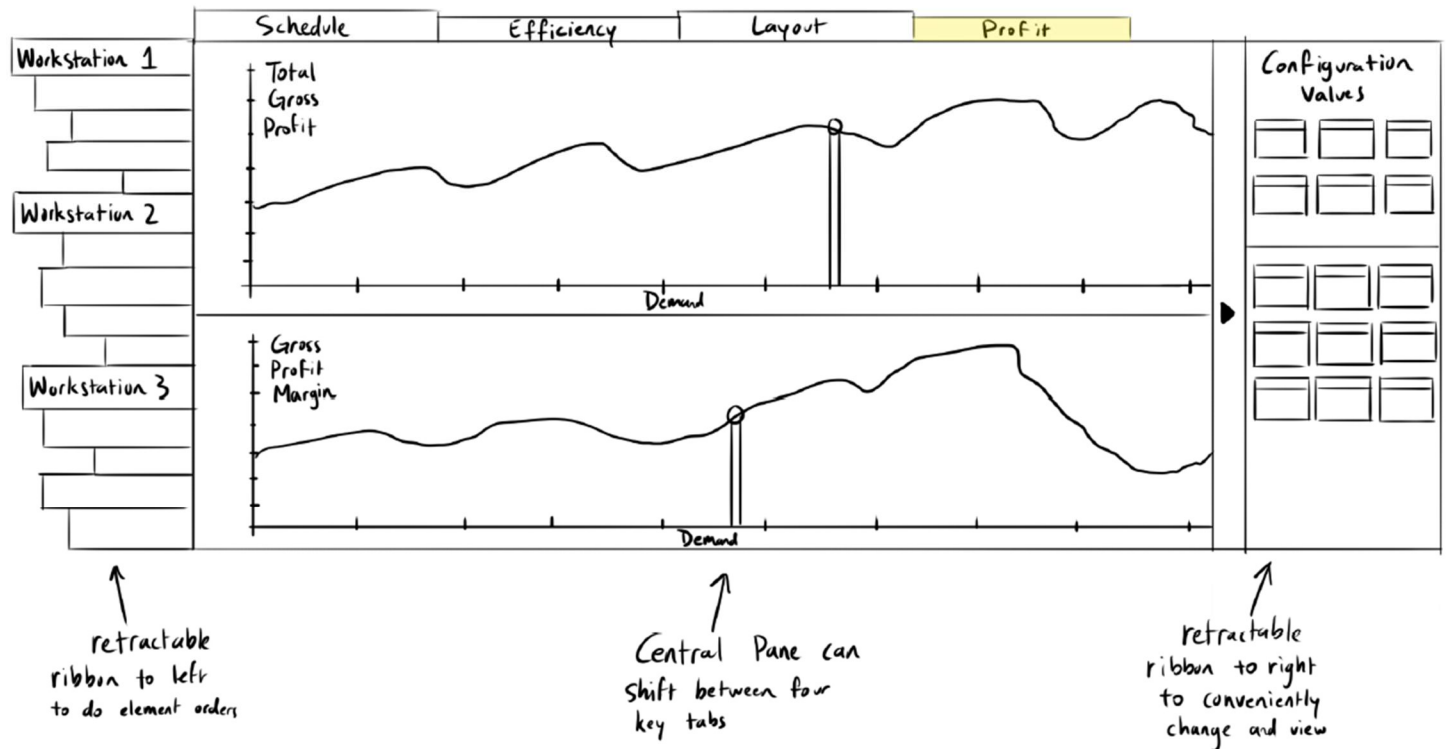
Factory Flow To Fortune

Abstract / Problem Definition

Start Here:

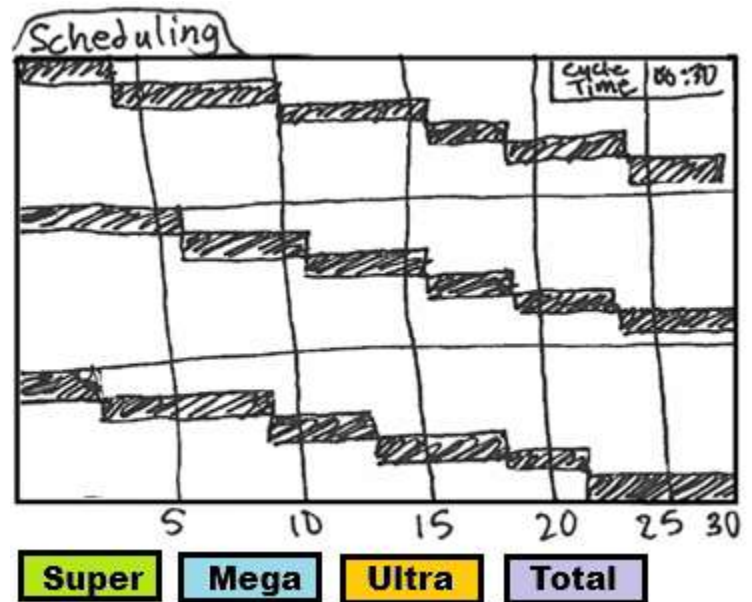
Man Power	Basic Financials	Business Expenditures						
<table border="1"><thead><tr><th>Number of Employees</th></tr></thead><tbody><tr><td></td></tr></tbody></table>	Number of Employees		<table border="1"><thead><tr><th>Demand</th></tr></thead><tbody><tr><td></td></tr></tbody></table>	Demand		<table border="1"><thead><tr><th>Labor Cost</th></tr></thead><tbody><tr><td></td></tr></tbody></table>	Labor Cost	
Number of Employees								
Demand								
Labor Cost								
<table border="1"><thead><tr><th>Operational Hours</th></tr></thead><tbody><tr><td></td></tr></tbody></table>	Operational Hours		<table border="1"><thead><tr><th>Sell Price</th></tr></thead><tbody><tr><td></td></tr></tbody></table>	Sell Price		<table border="1"><thead><tr><th>Cost of Goods</th></tr></thead><tbody><tr><td></td></tr></tbody></table>	Cost of Goods	
Operational Hours								
Sell Price								
Cost of Goods								

For the second page, a mixture of two variable ribbons will be used to shift configurations of the assembly line, with four different key visuals being seen in the middle using a tab system.



FINAL PROPOSED DASHBOARDS

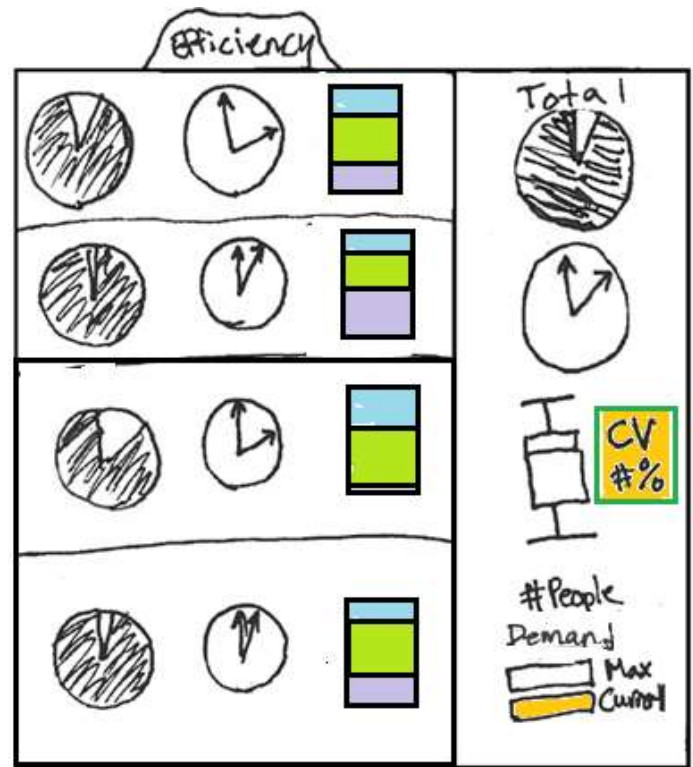
Scheduling: There is the issue of the original proposed schedule dashboard failing to show true scheduling, in that an operator will not spend all their time finishing one task before moving to the next. An alternative to fix this would have been a full simulation mode in which each of the three models flow through it with their respective tasks either shrinking or expanding the workstation. However, this dynamic view will grow increasingly complicated as the build sequence is fully explored. As while the belt movements will be constant, the employee movements will not. A compromise



which allows for true scheduling to be shown, while not being overly complex in its implementation on the backend, will be to allow a schedule per model to be shown. This will allow you to click on one of the three models (or total tasks) in the bottom left (as it will be consistently unobstructed) and it will show a Gantt Chart which will show the relevant tasks for each workstation in their element times (or in the case of Total, the daily element times) sequentially. If the true variation of the build sequence can be accomplished algorithmically, then a true simulation across builds will be explored. This will involve both a calculation of how long an operator is preoccupied with their allotted tasks, as well as when the next units are available. Especially for a full day build sequence, this becomes very complex, with a need to define if interstation movements can occur. A Gantt chart is a natural and effective way

to communicate both sequencing and the amount of time tasks will take within a greater timeframe. This will function as a visual extension of the left ribbon which will show a by model projection of tasks based on the given model. Finally, as the time each station spends on different models will vary, a cycle time display will be shown at the top right, showing the maximum time of the workstations below. This will be the initial scope for this dashboard, with a potential to expand it.

Efficiency: This will show a per workstation view of both proportion of efficient and idle times, as well as the discrete daily idle time, as well as what percentage of different builds make up the station's build times. To the right, it will show the comprehensive line values for this, as well as visualizations for how well balanced the line is between the different stations. A pie chart is an intuitive way to show the balance between the two opposing principles of idleness and efficiency; with a key goal of balancing a line to get each workstation fully filled up with efficient time, this quickly helps visualize how effective this balancing is. However, a discrete value of time is also useful, especially in evaluating the time-value of money. We will therefore show a clock which articulates the wasted time within a day, with a tooltip showing the dollar amount of that unapplied labor. To the side of that,



we will use a stacked-bar chart showing the ratio of how much of the daily work time of the station is spent on each of the three models. A stacked bar chart will give a visual contrast to the other circular charts. To the far right of this table, the full assembly line will be explored, using the pie-chart and block designs from before to show totals for the full line based on efficiency. Beyond simply reducing idle time, it is also desirable to reduce variance between stations, so we will use a standard box/whisker chart to show the distributions of the station allocations. However, a shortcoming of using these sorts of charts is that they can be misleading in how they compare different groups which have different meanings. We will therefore also show a Coefficient of Variance as a number to the side, with it being highlighted from red to green based on how good the CV for the allotment is. Below this, will be a display which shows the number of employees currently assigned, and below that, the maximum demand possible by that number of employees, in comparison to the current demand supported. This will similarly use a Red-Yellow-Green highlighting to show good values; this being more direct to understand for our target audience than using an actual scale visual.

Layout: Depending on if this can be designed as a simulation or not will impact greatly what this visual can accomplish, but our goal is for this to display the floor space allocation of the line and show the movement of different models across it into finished goods. It will also show visuals of key assembly line characteristics, such as conveyor speed and throughput. The use of this can communicate both floor space (through grid lines), sequencing of both elements and also models built, the speed of movement through the assembly line, both as cycle time and conveyor speed, as well as the WIP supported by the line at a given time, as well as the key principle that a U-shape is often optimal for an assembly line's operator flow (minimizing movements from the beginning and the end). While this

simulation effectively will communicate these many variables, it would be supported by an explicit description of these variables on the side. Speed is meaningfully communicated with a speedometer, so we could use that to show how quickly the line must move, and around it we could display both Throughput and WIP as a numerical output. If an active simulation of can be fully designed, a bucketed chart showing the accumulation of finished goods from the three models would be a good way to communicate how the behavior of the line over time behaves as expected from a cumulative perspective. It is believed that this will be the hardest visual to design; therefore, it will be a focus for us during fall break, when we can focus on it. This final design will be determined based on feasibility of implementation, and for now will aim to be similar to the prior shown design.

Profit: This is intended to show the profitability of the assembly line based on different demand levels, if the elements can be configured to move around, then we will also show optimal and current profitability within this chart as well. While there are more complex financial metrics we could show, the scope of this project is to teach about the functionality of assembly lines, rather than financial ratios. To make this meaningful from a business perspective, we would need to greatly expand out input variables, which will double the complexity of our landing page, and thus make for a less streamlined experience. Ideally, it would be good to show break-even points, but that would involve factors such as an understanding of the microeconomics of the refrigerator market, as well as the cost of different assembly line components. The reality is that purchasing turns on an assembly line does cost more money, despite allowing for more efficient operator flow. But this would add a complexity to this which, while not difficult to program, would be confusing for users of the visualization. It is unlikely that they will be familiar with both factory physics and also financial planning. Therefore, a naïve understanding of gross profit and its margins will be sufficient for the purposes of determining optimal design—although it is more nuanced than that. The most important variable for determining a configuration of an assembly line is the demand for the product, and the most important variable for determining if a configuration is good is going to be how profitable it is. In this way, the most valuable macro-relationship that someone should be able to visualize across all configurations is the relationship between demand and profitability. While the overall complexity of interactions between variables makes it difficult to compare them in a collected visual, but rather through a series of dashboards, this relationship can be shown across all viable values of demand. By using a slider which moves along the “optimal” trend-line between demand and profit, we can also use a vertical slider at that point to communicate what the current configuration’s profitability is. This will allow someone to view how optimal their current configuration is at making money. The use of tooltips can be used to convey the adjustable variables, so that the user can see how their current selection aligns with the optimal configuration for their selected demand—as well as to see if it is financially viable to increase that demand. The current plan for this is currently seen in the second page of our proposed final templating.

Sequencing (Stretch Goal): This is not currently planned with the given scheduling, but if development ends up moving smoothly, we may also do a fifth tab which shows the Build Sequencing and Bayesian Weights. While a roughly optimal balanced sequence is shown under our configurations here, a change in demand will actually slightly change this. Therefore, an authentic depiction will require an algorithmic implementation, which while much simpler than those used to generate optimal sequence so elements, it will still require some extensive work on the backend. If time permits this algorithmic development, then this would be a helpful visualization. However, it has lower value compared to implementation complexity than the other tabs and will not be planned on currently.

Must-Have Features

- Ability to update adjustable variables to alter output variables
- Pert Chart to communicate precedence of assembly steps
- Ability to show 10 different assembly line configurations, worker counts
- Showing both element times and total workload times
- Both a per workstation and total line visualization of efficiency and balance of the line
- Visual depiction of the floor space of the 10 base configurations
- A landing page introducing the problem-space, and getting initial adjustable variables
- Show Scheduling based on Model Built
- Ability to show Profit across all acceptable demand levels

Optional Features

- Real-Time updating of variables
- Allowing active switching order of elements in workstations
- Ability to show current value vs optimal value for financials
- Full Animation of the Assembly line in layout tab
- Retractable Ribbons and tabs for variables and workstation.
- Show full simulation of Units Built In optimal build sequence
- An optimal build sequence tab which shows mixed assembly sequencing.

Project Schedule

5. Page Structure and Landing Page Designed
 - a. Joel – Write the Landing Page Descriptions
 - b. Sumaiya – Design Inputs and general variable structures
 - c. Dhruv – CSS and layout
6. Page 2 page designed, Ribbons (Left, then Right)
 - a. Joel – Backend Development
 - b. Sumaiya – PERT
 - c. Dhruv – CSS on 2nd Page, Right Ribbon
7. Layout Tab (get a physical depiction of line) - Collaborative Brainstorming/Work over break
8. Left Ribbon, Webpage Skeleton (make sure basic links work)
 - a. Joel – Left Ribbon and residual backend
 - b. Dhruv – Left Ribbon
 - c. Sumaiya – Clean-up Website and test links
9. Milestone – catch up on anything missed before
10. Scheduling and Efficiency Tab (contents)
 - a. Joel – Support on both Tabs
 - b. Dhruv – Scheduling
 - c. Sumaiya - Efficiency
11. Profit Tab, Sequencing Tab (optional), and Layout Cleanup (contents) - TBD
12. Bug Fixing – Full Team Collaboration
13. Screencast Submission
14. Revisions – Full Team Collaboration
15. Project Due